

Previously:

• Turán thm $ex(n, K_{r+1}) \approx \left(1 - \frac{1}{r}\right) \frac{n^2}{2}$

• Erdős-Simonovits Stability

$$G \begin{cases} K_{r+1}\text{-free} \\ e(G) \geq ex(n, K_{r+1}) - o(n^2) \end{cases} \Rightarrow |e(G) - T_r(n)| = o(n^2)$$

• Motzkin-Straus

Today

Erdős-Simonovits-Stone $\forall H$

$$\Rightarrow ex(n, H) = \left(1 - \frac{1}{\chi(H)-1}\right) \frac{n^2}{2} + o(n^2) \Leftarrow$$

Rmk: if $\chi(H) \geq 3$, $\uparrow$ coeff. $\neq 0$

$$\chi(H)=4$$

pf of the lower bd: Take $T_{\chi(H)-1}(n)$



Notation edge density of G is $\frac{e(G)}{\binom{n}{2}} = p$

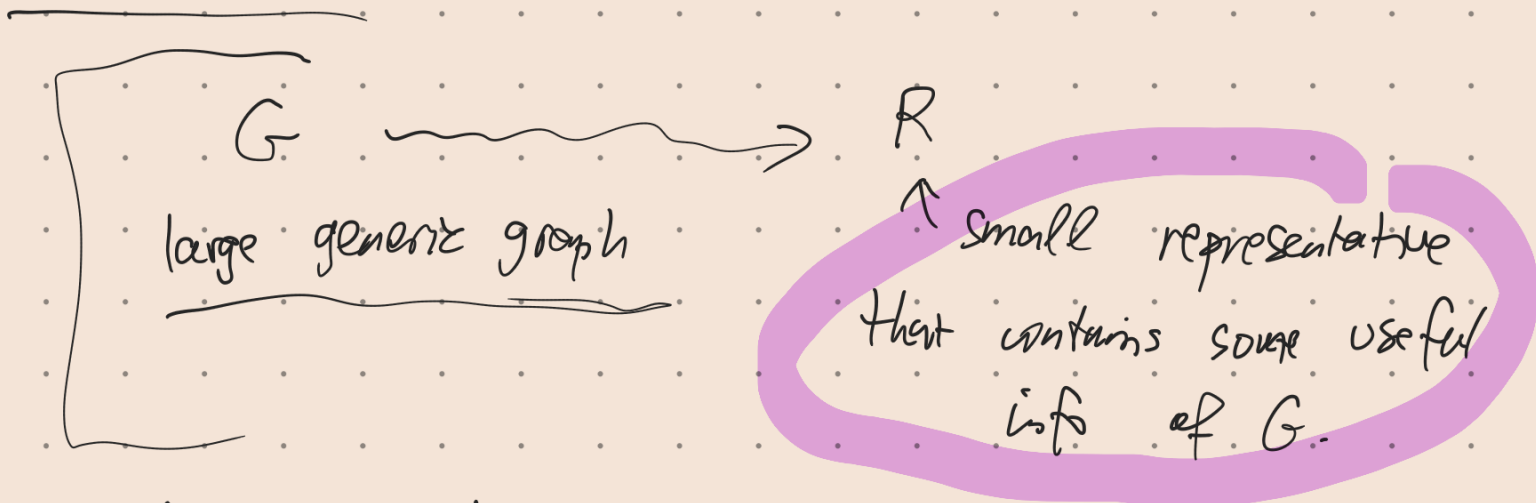
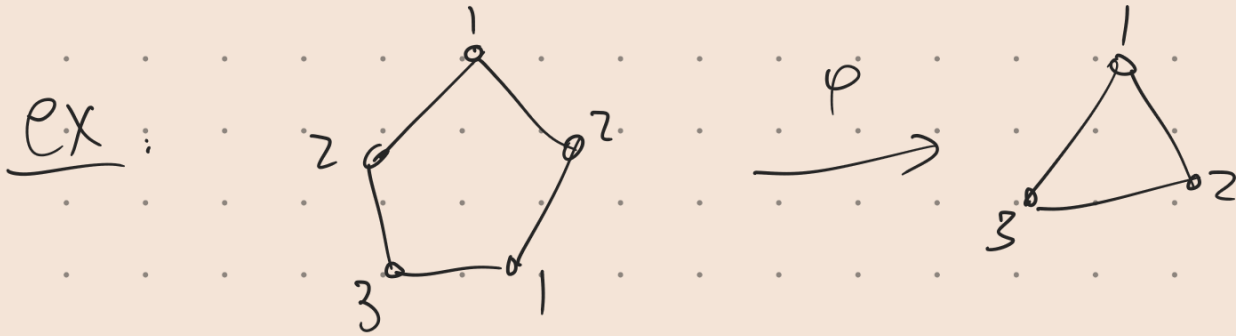
A homomorphism from H to F

$\varphi: V(H) \rightarrow V(F)$ is a map

preserving adjacencies. That is

$$\forall u \sim_H v \Rightarrow \varphi(u) \sim_F \varphi(v)$$

- We say in this case that F is a homomorphic image of H .



Lem (Embedding Lem) Let H be a graph. For large n ,

$$\begin{aligned} & \bullet G \begin{cases} \text{size } n \\ H\text{-free} \\ \text{edge density } \tau \end{cases} \rightsquigarrow \exists R \begin{cases} H\text{-free} \\ e(R) \geq (\tau - o(1)) \binom{|R|}{2} \end{cases} \\ & \hspace{15em} \text{reduced graph} \end{aligned}$$

Pf upper bound of ESS: Let $\chi(H) = r$.

- K_r is a homomorphic image of H .

$\Rightarrow R$ is K_r -free as G is H -free

• Turán's thm $\Rightarrow (1 - \frac{1}{r+1}) \binom{|R|}{2} \leq e(R) \leq (1 - \frac{1}{r+1}) \binom{|R|}{2}$ ~~to (R)~~

Stability method

• Step 1. Obtain asymptotic result

• Step 2. Obtain stability $\Leftarrow$

• Step 3. Use stability to get exact result.

$\uparrow$ Stability method

Thm $ex(n, C_5) = \lfloor \frac{n^2}{4} \rfloor$

Lem (Stability for C_5) Let $\epsilon > 0$, $\exists \delta > 0$ s.t.

for large n :

$$G \begin{cases} n\text{-vx } C_5\text{-free} \\ e(G) \geq \frac{n^2}{4} - \delta n^2 \end{cases} \Rightarrow$$

G can be made bipartite by deleting $\leq \epsilon n^2$ edges.

pf (C_5 upper bd) Let G be extremal C_5 -free graph. $\Rightarrow e(G) \geq \lfloor \frac{n^2}{4} \rfloor$ $\leftarrow$ ~~(scribble)~~

Claim $\exists G' \subseteq G$ w./ $\begin{cases} |G'| \geq \frac{n}{2} \\ \delta(G') \geq (\frac{1}{2} - \epsilon) |G'| \end{cases}$

Idea keep removing low-deg v's

$\leadsto$ if $|G'| = \frac{n}{2}$

$$\Rightarrow e(G') \geq (\frac{1}{4} + \epsilon') |G'|^2$$

$\hookrightarrow$ to ESS. ~~min~~ min deg

Assume now $\delta(G) \geq (\frac{1}{2} - \epsilon) n$

Take a max-cut $X \cup Y$. $|X| \sim \frac{n}{2} \sim |Y|$

Suffices to show that

$$e_G(X) = e_G(Y) = 0$$

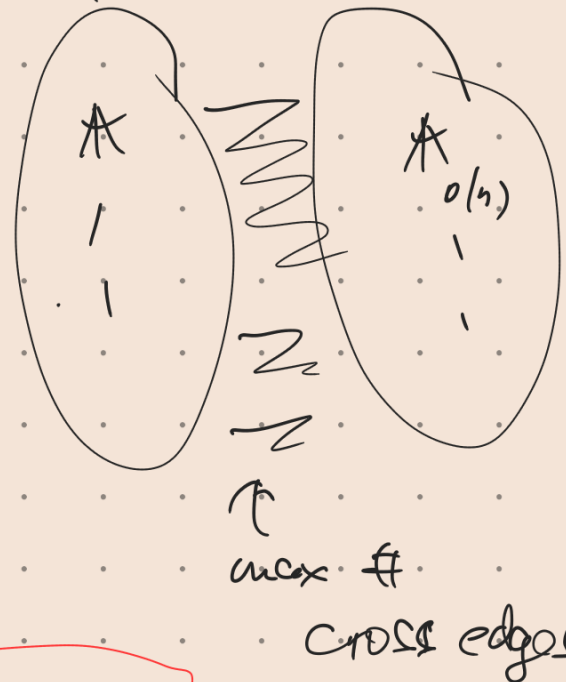
$$\bullet \quad e_G(X) + e_G(Y) \leq \epsilon n^2 \Leftarrow$$

by stability of G

$$\leadsto |X|, |Y| = \frac{n}{2} \pm 2\sqrt{\epsilon} n$$

for otherwise $e(G) \leq |X||Y| + \epsilon n^2 < \frac{n^2}{4} \hookrightarrow$

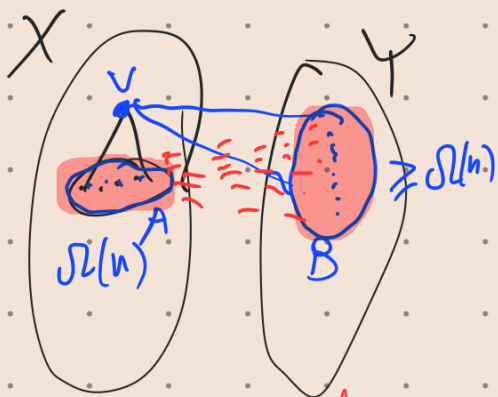
claim $\Delta(G[X]), \Delta(G[Y]) \leq 2\sqrt{\epsilon} n$



pf: $[A, B]$ is $\mathcal{P}_4$ -free

$$\Rightarrow e(A, B) = O(n)$$

$$\leadsto e(G) < \frac{n^2}{4}$$



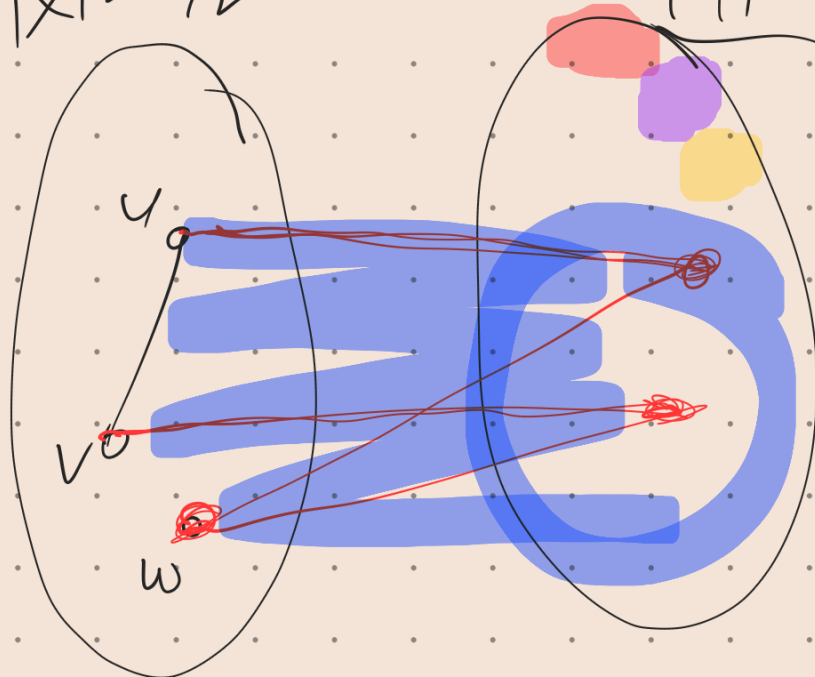
$\Omega(n^2)$ edges missing

Suppose X has an edge

$$|X| \approx \frac{n}{2}$$

$$|Y| \approx \frac{n}{2}$$

$$e(G) \geq (\frac{1}{2} - \epsilon)n$$



Bipartite Turán Problem

$\forall H$ bipartite

$$\bullet \text{ ESS} \Rightarrow ex(n, H) = o(n^2)$$

$$\bullet ex(n, H) = O(n^{2-c_H}) \dots (*)$$

< Füredi - Simonovits Survey >

Thm $ex(n, C_4) \leq \left(\frac{1}{2} + o(1)\right) n^{3/2}$

pf • double-counting Chernie's



Let G be C_4 -free n -vx graph.

$$\# \text{ Chernie's} = \sum_{v \in V(G)} \binom{d(v)}{2}$$



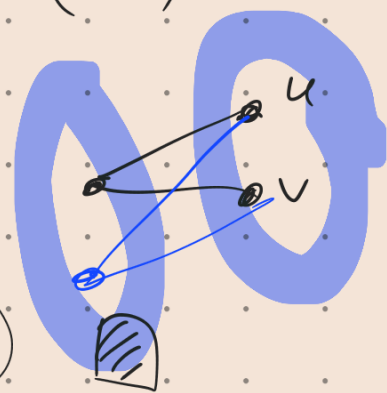
Jensen's ineq

$$\geq n \cdot \binom{\frac{1}{n} \sum d(w)}{2} = n \binom{\frac{1}{n} 2e(G)}{2} \dots (LB)$$

$$\# \text{ Chernie's} \leq \binom{n}{2} \dots (UB)$$

as G is C_4 -free

$$(LB) \leq (UB) \dots \dots \dots \text{smiley face}$$



Note that $C_4 = K_{2,2}$

Thm (Kővari - Sós - Turán)

$2 - \frac{1}{s}$

(Exercise) $\forall s \leq t, ex(n, K_{s,t}) \leq t \cdot n$

History No matching LB of $\Omega(n^{2-k})$

Cases known $\left\{ \begin{array}{l} S = 2, 3 \\ \text{or} \\ t \geq (S-1)! + 1 \end{array} \right.$

Open Improve $\Omega(n^{5/3}) \leq \text{ex}(n, K_{4,4}) \leq O(n^{7/4})$

— Connection to Sidon set —

$$\text{ex}(n, C_4) = \Theta(n^{3/2})$$

Def A **Sidon set** is a set $\subseteq \mathbb{N}$

with no non-trivial solⁿ to

$$a + b = c + d$$

Known $A \subseteq [n]$ Sidon $\Rightarrow$ $|A| = (1+o(1))\sqrt{n}$
max-size

Ex • Show $\nexists$ Sidon set $A \subseteq [n]$

$$|A| \leq \sqrt{2n}$$



Use a Sidon set of size $\Theta(\sqrt{n})$
to construct C_4 -free n -v graph w.
 $\Theta(n^{3/2})$ edges.

Thm (Erdős-Rényi-Sós)

$$ex(n, C_4) \geq \left(\frac{1}{2} - o(1)\right) n^{3/2}$$

Pf: $\exists$ prime p s.t.

$$(1-o(1))\sqrt{n+1} \leq p \leq \sqrt{n+1} \quad (p^2 \approx n)$$

$$G \begin{cases} V(G) = \mathbb{F}_p^2 \setminus \{(0,0)\} & \#V = p^2 - 1 \\ E(G): (a,b) \sim (x,y) \text{ if } ax+by=1 \end{cases}$$

C_4 -free:

Take two v 's v_1, v_2

$$v_i = (a_i, b_i)$$

$$\forall i \in [2]$$

$N(v_i)$ all (x,y) w/ $a_i x + b_i y = 1$

a line!

$\Rightarrow v_1, v_2$ has ≤ 1

common neighbour $\Rightarrow C_4$ free.

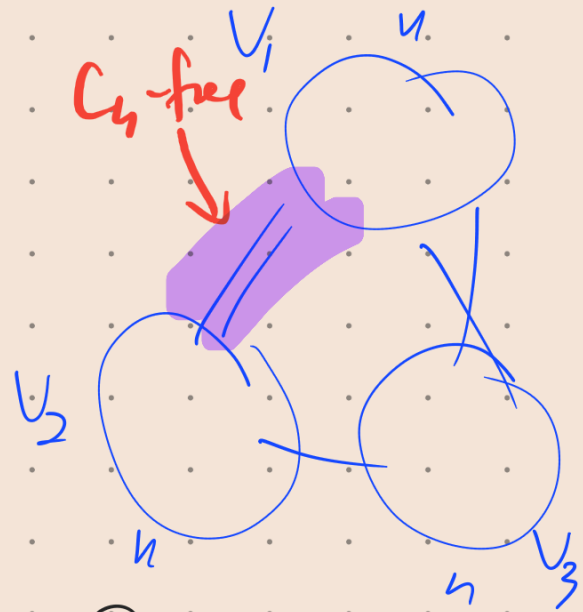
every v has $\deg = \frac{p}{p-1} \dots$



Open problem (Fischer & Matousek)

Let G be $n \times n \times n$ 3-part.

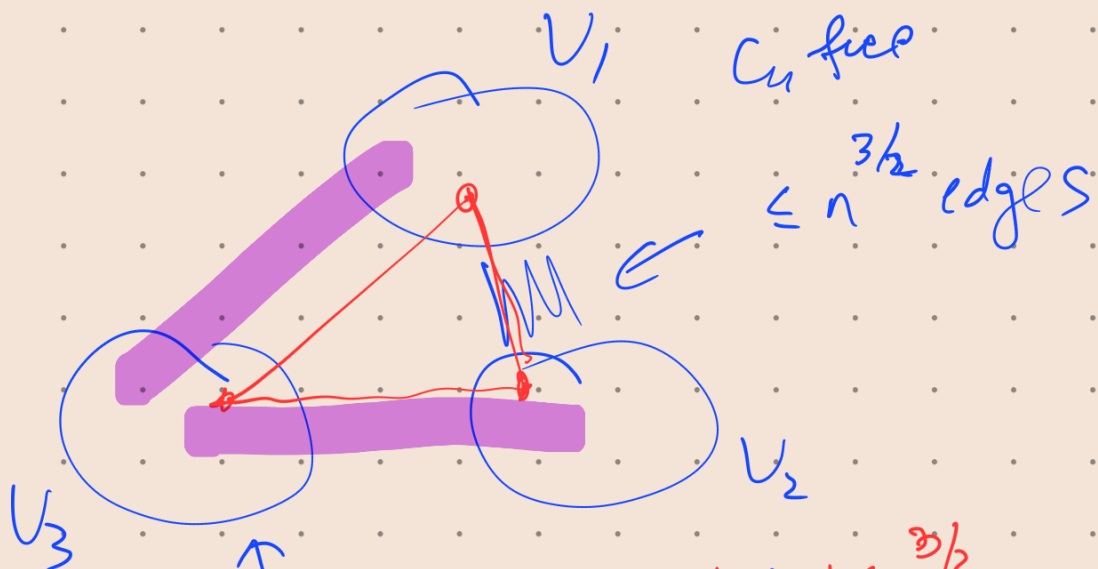
$[V_i, V_j]$ C_4 -free



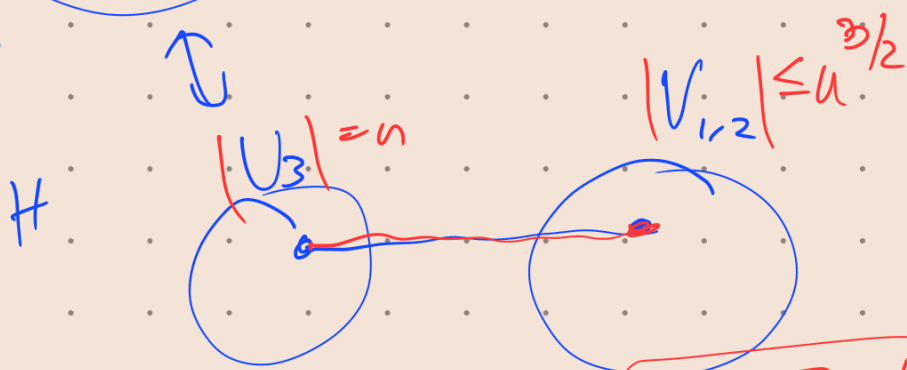
Q: What is max. #
 Δ such that G can have?

Known

$$\underline{n^{5/3}} \leq \Delta \leq n^{7/4}$$



C_4 -free
 $\leq n^{3/2}$ edges



$|V_{1,2}| \leq n^{3/2}$ a u, v in $V_{1,2}$ ~ edge in (V_i, V_j)

H is also C_4 -free $\leq n^{7/4}$